HCI Research as Problem Solving

There are three types of research problems in HCI: empirical, conceptual, and constructive. Each of these categories has three subtypes. Empirical problems are defined as those aimed at creating or describing real-world phenomena related to the use of computing. Laudan, who developed the concepts of problems and solutions as foundational to science, cites three characteristics of empirical problems: unknown phenomena, unknown factors, and unknown effects.

Conceptual problems are non-empirical and involve issues in theory development in the broadest sense. They are also what Laudan calls second-order problems. Work on conceptual research problems aims at explaining previously unconnected phenomena in interactions. Laudan defines three characteristics of conceptual problems: implausibility, inconsistency, and incompatibility.

Constructive research is focused on producing understanding about constructing interactive artifacts for human use of computing. These problems are characterized by (1) no known solution, (2) partial, ineffective, or inefficient solutions, and (3) insufficient knowledge or resources for implementation or deployment.

Laudan introduced the concept of **problem-solving capacity**, which refers to how well a solution resolves significant problems. He argues that it is more important to evaluate how solutions improve our capacity to address research problems than whether the solutions are strictly 'true' or 'well-confirmed'.

There are five problem-solving capacities in HCI research:

1. **Significance** – A solution addresses a problem that is important to stakeholders, whether they are researchers, practitioners, or end users.

2. **Effectiveness** – The solution resolves the essential aspects of the stated problem.

3. **Efficiency** – This refers to the costs of applying a solution relative to the gains achieved.

4. **Transfer** – How well the solution transfers to neighboring problems or other instances of the problem.

5. **Confidence** – The probability that the proposed solution holds.

A research problem in HCI is a stated lack of understanding about some phenomenon in human use of computing, or a stated inability to construct interactive technology to address that phenomenon for desired ends.

HCI not only has a high tolerance for risk but also places strong emphasis on practical improvement potential, responding to issues of stakeholder groups and global problems like inequality. While Best Papers are a curated and special sample, unrepresentative of HCI as a whole, they provide insight into where the field may be heading and prove that there are multiple forms of progress. Problem-solving capacities in HCI are diverse and under-defined.